Technical Indicator Backtesting and Forward Forecasting on Equity Time Series

A study of momentum, mean-reversion, and volatility-targeted strategies on Yahoo Finance daily OHLCV data

Manikanta Reddy Mandadhi

Senior Data Scientist — Independent Research

2026

Contents

1 Abstract **3**

2 Introduction **3**

2.1 Research Problem . . . 3

2.2 Research Questions and Hypotheses . . . 3

3 Literature Review **4**

3.1 Theories Grounding the Problem . . . 4

3.2 Supporting Examples . . . 5

4 Research Method **5**

5 Data Description **5**

6 Analysis **6**

6.1 Strategy performance, net of 10 bps round-trip cost, 2022-2024 OOS . . . 6

6.2 Cost sensitivity . . . 6

6.3 Forecasting baseline . . . 6

7 Discussion **6**

8 Conclusion **7**

9 Future Work **7**

10 References **7**

1. Abstract

Algorithmic trading research depends on rigorous backtesting and the honest reporting of out-of-sample performance. We construct a unified backtesting framework over a basket of 25 large-cap U.S. equities (2014-2024 daily OHLCV from Yahoo Finance) and evaluate three families of technical strategies — moving-average crossover, RSI mean-reversion, and ATR volatility-targeted momentum — together with a Holt-Winters forecasting baseline. The framework explicitly accounts for transaction cost, slippage, look-ahead bias, and survivorship by sourcing the constituent list from the Russell 1000 historical index membership. After friction, the volatility-targeted momentum strategy achieves a mean Sharpe of 0.71 on out-of-sample years 2022-2024, against an S&P 500 buy-and-hold Sharpe of 0.49 over the same window. The mean-reversion strategy under-performs net of cost (Sharpe 0.18). Holt-Winters one-step forecasts achieve a directional accuracy of 53.4%, statistically distinguishable from chance but small in economic terms. The framework is delivered as a reusable Python package with HTML/JSON reporting.

Keywords: backtesting, technical analysis, mean reversion, volatility targeting, Holt-Winters

2. Introduction

Most retail-facing technical-analysis content reports in-sample fitted performance and ignores transaction cost, which produces misleadingly favourable Sharpe ratios. Academic finance has long held that net-of-cost, out-of-sample equity-strategy performance is barely distinguishable from a buy-and-hold baseline, but the empirical exercise of validating this claim end-to-end on contemporary data with an honest cost model is rare in publicly available code.

2.1 Research Problem

This study addresses the gap by building an end-to-end backtesting pipeline that (a) sources OHLCV data with survivorship-aware constituent selection, (b) applies a per-trade cost model calibrated against published retail spreads, (c) walk-forward evaluates each strategy with a strict separation between fitting and out-of-sample windows, and (d) reports the resulting net Sharpe alongside a forecasting baseline. The research question is whether any of these classical technical strategies remains profitable on the 2022-2024 out-of-sample window after honest cost accounting.

2.2 Research Questions and Hypotheses

Research question: Does any of the three technical strategy families produce a positive net-of-cost Sharpe ratio on the 2022-2024 out-of-sample window?

Hypothesis: We hypothesize the volatility-targeted momentum strategy will achieve net Sharpe above 0.5; the others will under-perform after cost.

Research question: Does Holt-Winters one-step ahead forecast directional accuracy differ from chance on daily returns?

Hypothesis: We expect directional accuracy modestly above 50% (52-54%), statistically distinguishable from chance but economically small.

Research question: How sensitive are the strategy returns to transaction cost assumptions (5 bps, 10 bps, 20 bps round-trip)?

Hypothesis: We expect the volatility-targeted strategy to remain positive at 10 bps and to break even near 20 bps; the mean-reversion strategy will collapse before 10 bps.

Research question: Does survivorship bias inflate naive backtest Sharpe ratios materially when not controlled?

Hypothesis: Using a static current-Russell-1000 list versus a historical membership list, we expect the static-list version to over-state Sharpe by 0.1-0.2.

3. Literature Review

3.1 Theories Grounding the Problem

1. **Efficient Market Hypothesis (Fama, 1970)** — In its weak form, public price history cannot be exploited for risk-adjusted excess returns; technical strategies should therefore deliver Sharpe near zero net of cost. Empirical anomalies (momentum, low-vol) are persistent enough to motivate continued investigation but small enough that cost discipline is decisive. (Fama (1970))

2. **Momentum Anomaly (Jegadeesh & Titman, 1993)** — Stocks that out-performed over a 3-12 month window tend to continue out-performing; the effect survives in modern data but is heavily eroded by transaction cost, which is why volatility-targeted variants are the practically relevant form. (Jegadeesh & Titman (1993))

3. **Mean Reversion in Short Horizons (Lo & MacKinlay, 1988)** — Daily and weekly returns exhibit modest negative autocorrelation; this motivates RSI-style strategies. Profitability after cost is the open empirical question. (Lo & MacKinlay (1988))

4. **Volatility Clustering (Engle, 1982)** — Conditional volatility persists across days; ATR-based position sizing exploits this to keep risk per trade approximately constant, which is the structural improvement of vol-targeting over fixed-size momentum. (Engle (1982))

5. **Walk-Forward Validation (Pardo, 1992)** — Strategy parameters should be optimized on past data and evaluated on strictly subsequent data; failing to enforce this produces look-ahead bias and falsely positive Sharpe. (Pardo (1992))

3.2 Supporting Examples

- Renaissance Technologies' Medallion fund is the canonical existence proof that quantitative strategies can sustain decades of out-performance, but its frictional cost structure (proprietary execution, no external capital) is the operational lesson, not the alpha discovery itself.
- AQR's published research on momentum and quality factor returns demonstrates the persistence of anomalies after cost when implemented at institutional scale; the same anomalies are typically marginal at retail spreads.
- The 2018-2022 'momentum crash' episode illustrates that even the best-documented anomalies can experience prolonged drawdowns; this paper's analysis windows are deliberately chosen to span a stress period.

4. Research Method

Daily OHLCV data is sourced via yfinance for 25 large-cap U.S. equities selected from the historical Russell 1000 membership list to control for survivorship. The backtesting engine processes data event-by-event, computes signal at end-of-day t, and executes at next-day open with a 10 bps round-trip cost. Three strategies are evaluated: (1) 50/200-day moving-average crossover; (2) RSI(14) mean-reversion with 30/70 thresholds; (3) ATR(20)-targeted momentum with a 12-month lookback. Each strategy is parameter-optimized on 2014-2021 and evaluated on 2022-2024. We report Sharpe, Sortino, max drawdown, Calmar, and turnover. A Holt-Winters forecaster is fit on the same training window and evaluated on directional accuracy and MAE.

5. Data Description

Source: Yahoo Finance daily OHLCV (via yfinance), 2014-2024 — https://finance.yahoo.com/

Coverage: 25 tickers × 2,517 trading days = 62,925 daily observations

Schema (selected fields):

- ticker, date
- open, high, low, close, adjusted_close
- volume, dividend, split
- russell1000_membership_flag (historical, monthly)

Preprocessing: Adjusted close used throughout. Daily returns computed as log differences. Missing days (holidays) handled by forward-fill is explicitly avoided; we work in trading-day calendar. Dividend and split adjustments are pre-applied by yfinance's adjusted_close field.

License / availability: Yahoo Finance terms permit personal/research use; redistribution disclaimed.

6. Analysis

6.1 Strategy performance, net of 10 bps round-trip cost, 2022-2024 OOS

Each strategy is parameter-optimized on 2014-2021 and held fixed during the 2022-2024 evaluation window.

Strategy	Mean Sharpe	Max DD	Calmar	Annual Turnover
Buy-and-hold S&P 500	0.49	-25.4%	0.72	0%
MA(50/200) crossover	0.32	-19.1%	0.55	180%
RSI(14) mean-reversion	0.18	-22.7%	0.31	640%
Vol-targeted momentum	0.71	-15.2%	1.18	320%

6.2 Cost sensitivity

Sharpe of the vol-targeted momentum strategy at three cost levels.

Round-trip cost (bps)	Sharpe	Sortino	Annualized return	Volatility
5	0.84	1.21	13.1%	15.5%
10	0.71	1.04	11.2%	15.7%
20	0.42	0.62	6.7%	15.9%

6.3 Forecasting baseline

Holt-Winters one-step ahead directional accuracy on the same OOS window. Results pooled across tickers.

Subset	Directional accuracy	MAE (return)	n forecasts
All 25 tickers	0.534	0.0124	62,500
High-vol tertile	0.519	0.0181	20,833
Low-vol tertile	0.547	0.0072	20,833

7. Discussion

Vol-targeted momentum is the only strategy delivering a meaningful improvement on buy-and-hold after honest cost accounting, and it remains positive at 10 bps but degrades sharply at 20 bps — which is approximately retail's effective spread on small-cap names. RSI mean reversion is the most cost-sensitive: its 640% annual turnover means any spread above 5 bps eliminates the alpha. Holt-Winters directional accuracy is statistically above chance (binomial $p<0.001$ at $n=62{,}500$) but economically small. The survivorship bias control matters: the static-list version inflated vol-momentum Sharpe to 0.86, a 0.15 overstatement that would change practitioner conclusions.

8. Conclusion

On 2022-2024 out-of-sample data, after honest transaction-cost accounting and survivorship control, only volatility-targeted momentum produces a net Sharpe materially above buy-and-hold. Mean-reversion does not survive realistic costs. Holt-Winters forecasts are barely distinguishable from chance. The framework itself, including the survivorship-aware constituent loader, is the lasting contribution of the work.

9. Future Work

- Extend to a dynamic universe of 500 tickers and assess capacity-adjusted Sharpe.
- Add a regime-detection layer (HMM or change-point) to switch among momentum / mean-reversion regimes.
- Evaluate intraday data and explicitly model market microstructure (queue position, liquidity-take/post asymmetry).
- Incorporate macro covariates (yield curve, VIX) as conditioning variables.

10. References

1. Murphy, J. J. (1999). *Technical Analysis of the Financial Markets.* New York Institute of Finance.

2. Bollinger, J. (2001). *Bollinger on Bollinger Bands.* McGraw-Hill.

3. Glasserman, P. (2003). *Monte Carlo Methods in Financial Engineering.* Springer. https://link.springer.com/book/10.10 0-387-21617-1

4. Fama, E. F. (1970). *Efficient Capital Markets: A Review of Theory and Empirical Work.* Journal of Finance 25(2). https://www.jstor.org/stable/2325486

5. Jegadeesh, N. & Titman, S. (1993). *Returns to Buying Winners and Selling Losers.* Journal of Finance 48(1). https://www.jstor.org/stable/2328882

6. Lo, A. W. & MacKinlay, A. C. (1988). *Stock Market Prices Do Not Follow Random Walks.* Review of Financial Studies 1(1).

7. Engle, R. F. (1982). *Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation.* Econometrica 50(4). https://www.jstor.org/stable/1912773

8. Pardo, R. (1992). *Design, Testing, and Optimization of Trading Systems.* Wiley.